

SOLE SOURCE JUSTIFICATION FOR ACQUISITION UNDER THE SIMPLIFIED ACQUISITION THRESHOLD

1. Contracting Agency and Activity. The Department of the Interior, United States Geological Survey, Alaska Science Center.

2. Nature of Action Being Approved. A new firm, fixed price purchase order to GeoSep Services, LLC for apatite fission-track (AFT) and uranium-lead analyses from geological samples.

3. Description of Supplies or Services. Fission-track and uranium-lead analyses of sedimentary rock samples is needed to evaluate depositional ages, sediment routing, and exhumation on Kuskokwim mountains (Km) region, Alaska, to understand the history of faults involved in producing large earthquakes and tsunamis and to understand the geologic framework of this region.

4. Estimated Dollar Value.

5. Statutory Authority. The proposed action may be awarded on a sole source basis under the authority of FAR 13.106-1(b)(1) – only one source reasonably available.

6. Rationale Supporting Use of Citation in No. 5. AFT+U-Pb thermochronology requires specialty instrumentation and expertise. (GSS) provides quality mineral separations and apatite and zircon fission track and U-Pb analytical services. GSS is the only contractor to provide combined fission-track age dating services and U-Pb age information. Based on research and to our knowledge, no other labs can provide all of the requested services, and none offer ^{252}Cf irradiation. This method is critical to robust measurements of fission-track length that especially constrain potentially small variations in exhumation. Thus, only Geosep Services can meet the technical requirements.

7. Other Information. N/A.

8. The Efforts to Identify Additional Sources Including the Market Research Conducted. Sources Sought 140G0326Q0047 was posted on SAM.gov, and no vendors responded.

9. Future Plans to Permit Competition. Any future requirements will undergo market research and will be competed at the maximum extent practicable.

10. Recommendation and Certification from Program Office. Based on the above, I recommend this acquisition be conducted on a limited source basis. I certify that technical data which form a basis for this justification, that are the responsibility of technical or requirements personnel are complete and accurate.

Research Geologist

Date

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[REDACTED]

11. Certifications from the Contracting Officer:

- a. The contracting officer will include a determination that the price to the Government is fair and reasonable by comparing current market values for similar items.
- b. This justification is accurate and complete to the best of my knowledge and belief.

[REDACTED]

Yangzhi Deng, Contracting Officer

Date

12. Approval.

[REDACTED]

[REDACTED], Bureau Competition Advocate, acting

Date